

Application Exercise: Practice with Binomial and Poisson

Solution

A Clinical Trial: Binomial Example

Suppose 5 people receive a new vaccine. From clinical trial data, the probability that any individual experiences a mild side effect (e.g., sore arm, mild fever) is 0.2.

Let X be the number of people (out of 5) who experience a mild side effect.

Then $X \sim \text{Binomial}(n = 5, p = 0.2)$.

Exercise 1

What is the probability that exactly 2 out of 5 people experience a mild side effect? Use the `dbinom` function to compute your answer and provide a 1-sentence conclusion.

```
dbinom(2,size=5,prob=0.2)
```

```
[1] 0.2048
```

So, there is about a 20.5% chance that exactly 2 people will have side effects.

Exercise 2

What is the probability that at least 1 out of 5 people experiences a mild side effect? (Hint: Use the complement rule: $P(X \geq 1) = 1 - P(X = 0)$). Use the `dbinom` function to compute your answer and provide a 1-sentence conclusion.

```
1-dbinom(0,size=5,prob=0.2)
```

```
[1] 0.67232
```

So, there is about a 67% chance that at least 1 person will have a mild side effect.

Hospital Visits: Poisson Example

Suppose that in a hospital emergency department, the average number of asthma-related visits per hour is 3. Let Y be the number of asthma visits in one hour. Then $Y \sim \text{Poisson}(\lambda = 3)$

Exercise 3

What is the probability that in a given hour, exactly 5 asthma patients arrive? Use the `dpois` function to compute your answer and provide a 1-sentence conclusion.

```
dpois(5, lambda = 3)
```

```
[1] 0.1008188
```

So, there is about a 10% chance of exactly 5 asthma patients in one hour.